

CV

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Namhyun Kim

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 [Website](#)  [Google Scholar](#)

Research Interests

Beyond-5G/6G communication systems; massive MIMO; integrated sensing and communication (ISAC); AI/ML-based communications; wireless foundation models; semantic communications; Bayesian estimation.

Education

Arizona State University (ASU), Tempe, AZ, USA Jan. 2025 – Present

Doctor of Philosophy in **Electrical, Computer and Energy Engineering**

Advisor: Prof. Ahmed Alkhateeb.

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea Mar. 2021 – Feb. 2023

Master of Science in **Electrical Engineering** Cumulative GPA: 3.81/4.30

Dissertation: Location-Aware Downlink Beamforming with Marginal Training Overhead for MmWave FDD Massive MIMO Systems.

Advisor: Prof. Junil Choi.

Yonsei University, Seoul, South Korea Mar. 2015 – Feb. 2021

Bachelor of Science in **Electrical Engineering** (High Honors) Cumulative GPA: 4.04/4.30

Professional Experience

InterDigital, Inc. Philadelphia, PA, USA

Ph.D. Intern, Wireless Lab June 2026 – Aug. 2026

- Calibrated integrated sensing and communication (ISAC) channel models based on 3GPP TR 38.901 for joint sensing and communication simulations.
- Developed a semantic communication framework incorporating 3GPP TR 38.901 spatial consistency procedures to capture channel evolution across user locations.

SK Telecom, Co., Ltd. Seoul, South Korea

Manager, LTE/5G Radio Access Network (RAN) Performance Improvement Group Jan. 2022 – Sep. 2023

- Performed LTE/5G L1/L2/L3 network planning, engineering, troubleshooting, and performance analysis.
- Conducted commercial verification testing of 64-TRX massive MIMO systems with Samsung Electronics, Co., Ltd.
- Used Python, SQL, Kubernetes, Linux, and MATLAB for network engineering and analysis.

Publications

Manuscripts under Revision and Preprints

- **N. Kim**, S. Alikhani, and A. Alkhateeb, “LWM-TF: A Wireless Foundation Model on 5G NR Time-Frequency Resource Grids,” *IEEE Journal of Selected Topics in Signal Processing (JSTSP)*, **under revision**.
- **N. Kim**, M. S. Abouamer, J. Park, P. Popovski, and A. Alkhateeb, “Digital Twin-Aided Prescreening for User Scheduling in MU-MIMO Downlink Systems,” arXiv preprint arXiv:2608.10364, 2026. [[arXiv](#)].

Journal Articles

- **N. Kim**, J. Han, J. Choi, A. Alkhateeb, C.-B. Chae, and J. Park, “Integrated Sensing and Communications in Downlink FDD MIMO Without CSI Feedback,” *IEEE Transactions on Wireless Communications*, 25:2984–3000, 2026. [[Link](#)].
- J. Han, **N. Kim**, and J. Park, “Reducing Latency by Eliminating CSIT Feedback: FDD Downlink MIMO Transmission for Internet-of-Things Communications,” *IEEE Internet of Things Journal*, 13(6):11301–11315, 2026. [[Link](#)].
- **N. Kim**, I. P. Roberts, and J. Park, “Splitting Messages in the Dark—Rate-Splitting Multiple Access for FDD Massive MIMO Without CSI Feedback,” *IEEE Transactions on Wireless Communications*, 24(4):3320–3332, 2025. [[Link](#)].

Conference Papers

- **N. Kim**, S. Alikhani, and A. Alkhateeb, “LWM-Spectro: A Foundation Model for Wireless Baseband Signal Spectrograms,” *IEEE International Conference on Communications (ICC)*, Glasgow, UK, pp. 1–6, 2026. [[Link](#)].
- **N. Kim** and J. Park, “Robust Precoding and Rate-Splitting Multiple Access for FDD Massive MIMO Without CSI Feedback,” *IEEE International Symposium on Information Theory (ISIT)*, Ann Arbor, MI, USA, pp. 1–6, 2025. [[Link](#)].
- **N. Kim**, J. Han, and J. Park, “Integrated Sensing and Communications in FDD MIMO Without CSI Feedback: Towards FDD MIMO ISAC,” *22nd International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt)*, Seoul, South Korea, pp. 132–137, 2024. [[Link](#)].

Projects

LTE Link-Level Simulation

Intelligent & Secure underwater CommunicaTion (InSeCT), KRIT

Seoul, South Korea
Sep. 2023 – Apr. 2024

- Implemented the signal processing chain in an OFDM receiver.
- Derived a post-equalization SINR model for CFO-impaired OFDM transmission.
- Linked post-equalization SINR to BICM capacity.

5G Shared Network Commercial Deployment

SK Telecom

Daejeon, South Korea
Sep. 2022 – Aug. 2023

- Expanded 5G coverage in Korea through shared-network deployment with KT and LGU.
- Integrated and optimized network equipment while addressing security concerns related to shared infrastructure.
- Managed traffic loads and coordinated RAN operations to maintain service quality.

Undergraduate Capstone Project

School of Electrical Engineering, Yonsei University

Seoul, South Korea
Mar. 2020 – Aug. 2020

- Completed a department-led capstone project during the COVID-19 pandemic.
- Developed a campus access control system combining RFID and temperature measurement sensors.

Scholarships & Awards

Global Korea Scholarship: Study Abroad Scholarship Program

National scholarship supporting graduate studies abroad.

NIIED, Korean Government
2025–2026

Graduate Student Academic Excellence Scholarship

Merit-based scholarship recognizing academic excellence.

Wooyang Foundation
2021–2022

Korean Government-funded Student

Master's studies fully funded by Korean government scholarships.

KAIST
2021–2022

Department Chair Commendation

Recognized for tutoring international students at Songdo International Campus.

Dept. of EE, Yonsei University
Feb. 2021

Academic Excellence Award

Semester Honors (Spring 2016, Fall 2018, Spring 2019), High Honors (Spring 2020), Highest Honors (Fall 2019).

Yonsei University

Graduation with Distinction

Graduated with High Honors, placing in the top 3% of electrical engineering graduates.

Yonsei University
Feb. 2021

Teaching and Mentoring Experience

Graduate Teaching Assistant

Course: EE528: Engineering Random Processes, KAIST

Daejeon, South Korea
Sep. 2021 – Feb. 2022

- Served as a teaching assistant for Prof. Junil Choi's Engineering Random Processes course.

Academic Tutor for International Students

EE Honor Society Program, Dept. of Electrical Engineering, Yonsei University

Songdo, Incheon, South Korea
Sep. 2019 – Feb. 2021

- Tutored 21 international students in Engineering Mathematics and Engineering Physics.
- Conducted tutoring sessions at students' request; most participants achieved strong grades.

Skills

- **Programming:** Python, C/C++, MATLAB, SQL.
- **Software:** Linux, Kubernetes.
- **Soft Skills:** Communication, adaptability, problem-solving.